

A

Case Study Report on

**Title of Case Study**

Submitted in partial fulfillment of the requirements for ASSIGNMENT – II of  
the course DATABASE MANAGEMENT SYSTEMS in IV SEMESTER for the  
Academic Year 2025-2026 for

**BACHELOR OF ENGINEERING**

**In**

**COMPUTER SCIENCE AND ENGINEERING**

**By**

**Kausar Jehan (160624733156)**  
**Umama Nooreen(160624733188)**

Under the Guidance of

**Dr. P. R. Anisha**

Associate Professor

Department of Computer Science and Engineering



**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**  
**STANLEY COLLEGE OF ENGINEERING & TECHNOLOGY FOR WOMEN**  
**(AUTONOMOUS)**

(Affiliated to Osmania University, Approved by AICTE and Accredited by NAAC)  
Hyderabad - 500 001, Telangana, India

**STANLEY COLLEGE OF ENGINEERING & TECHNOLOGY FOR WOMEN**  
(AUTONOMOUS)  
(Affiliated to Osmania University, Approved by AICTE and Accredited by NAAC)  
Hyderabad - 500 001, Telangana, India

**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**



**CERTIFICATE**

This is to certify that the Case Study report entitled, "**Traffic Violation Management System**", done by **Kausar Jehan, Umama Nooreen**, bearing roll numbers **160624733156, 160624733188** respectively, has been submitted to the Department of Computer Science and Engineering, **STANLEY COLLEGE OF ENGINEERING & TECHNOLOGY FOR WOMEN**, in partial fulfillment of the requirements for Assignment – II of the course Database Management Systems in the IV Semester of Bachelor of Engineering in Computer Science and Engineering for the Academic Year 2025-2026.

It is certified that they have completed the case study satisfactorily.

**Name & Signature of Faculty**

**Dr. P. R. Anisha**

Associate Professor CSE

**Signature of Head of the  
Department**

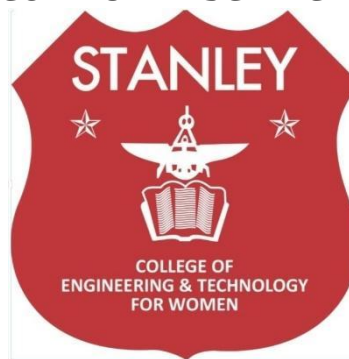
**Dr. R. Madana Mohana**

Professor & Head

**STANLEY COLLEGE OF ENGINEERING & TECHNOLOGY FOR WOMEN  
(AUTONOMOUS)**

(Affiliated to Osmania University, Approved by AICTE and Accredited by NAAC)  
Hyderabad - 500 001, Telangana, India

**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**



**DECLARATION**

We hereby declare that the work described in this report entitled “**Traffic Violation Management System**” which is being submitted by us in partial fulfillment of the requirements for **Assignment – II** of the course **Database Management Systems** in the **4<sup>th</sup> Semester** of **Bachelor of Engineering** in the **Department of Computer Science and Engineering, Stanley College of Engineering & Technology for Women** for the **Academic Year 2025-26**, is our original work.

The work is original and has not been submitted for any Degree or Diploma of this or any other university.

**Signature of the Student**

Kausar Jehan  
(160624733156)

**Signature of the Student**

Umama Nooreen  
(160624733188)

**STANLEY COLLEGE OF ENGINEERING & TECHNOLOGY FOR WOMEN**  
**(AUTONOMOUS)**

(Affiliated to Osmania University, Approved by AICTE and Accredited by NAAC)  
Hyderabad - 500 001, Telangana, India

**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**



**CASE STUDY DETAILS**

Course Code: SPC0403CS    Credits: 3    Evaluation: CIE - 5 Marks

**Course:** Database Management Systems

**Title:** Case Study Report on Traffic Violation Management System

**Name of the Students:** Kausar Jehan, Umama Nooreen

**Roll Numbers:** 160624733156, 160624733188

**Batch / Section:** CSE-C

**Faculty Supervisor:** Dr. P. R. Anisha

## RESUME OF THE TEAM MEMBERS

(Attach copies of your resume)



**STANLEY COLLEGE OF ENGINEERING & TECHNOLOGY FOR WOMEN**

(Autonomous)

Abids, Hyderabad - 500 001, Telangana

# *Department of Computer Science and Engineering*

## **Vision of the Institute**

Empowering girl students through professional education integrated with values and character to make an impact in the world.

## **Mission of the Institute**

- Providing quality engineering education for girl students to make them competent and confident to succeed in professional practice and advanced learning.
- Establish state-of-art-facilities and resources to facilitate world class education.
- Integrating qualities like humanity, social values, ethics, leadership in order to encourage contribution to society.

## **Vision of the Department**

Empowering girl students with the contemporary knowledge in computer science engineering for their success in life.

## **Mission of the Department**

- To impart quality education for girl students to learn and practice various hardware and software platforms prevalent in industry.
- To achieve self-sustainability and overall development through Research and Development activities.
- To provide education for life by focusing on the inculcation of human & moral values through an honest and scientific approach.
- To groom students with good attitude, team work and personality skills.

## **Program Educational Objectives (PEOs)**

**PEO1.** Graduates shall have enhanced skills and contemporary knowledge in software and hardware technologies for professional excellence, towards successful employment, advanced learning and research.

**PEO2.** Graduates shall have life-long learning attitude, innovation and creativity to master latest technologies, devise solutions for realistic and social issues in the society.

**PEO3.** Graduates shall have good attitude and personality skills, ethical values, teamwork and leadership skill towards professionalism and ethical practices within the organization and the society.



**STANLEY COLLEGE OF ENGINEERING & TECHNOLOGY FOR WOMEN**

(Autonomous)

Abids, Hyderabad - 500 001, Telangana

## *Department of Computer Science and Engineering*

### **Knowledge and Attitude Profile (WK)**

|             |                                                                                                                                                                                                                                                                                                  |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>WK1.</b> | A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.                                                                                                                                                         |
| <b>WK2.</b> | Conceptually-based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.                                                                                    |
| <b>WK3.</b> | A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.                                                                                                                                                                                       |
| <b>WK4.</b> | Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.                                                                                         |
| <b>WK5.</b> | Knowledge, including efficient resource use, environmental impacts, whole-life cost, reuse of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.                                                                              |
| <b>WK6.</b> | Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.                                                                                                                                                                                              |
| <b>WK7.</b> | Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development.                                                                            |
| <b>WK8.</b> | Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.                                                                                                        |
| <b>WK9.</b> | Ethics, inclusive behavior and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes. |



## STANLEY COLLEGE OF ENGINEERING & TECHNOLOGY FOR WOMEN

(Autonomous)

Abids, Hyderabad - 500 001, Telangana

# Department of Computer Science and Engineering

## Program Outcomes (POs)

|       |                                                                                                                                                                                                                                                                                                                        |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PO1.  | <b>Engineering Knowledge:</b> Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.                                                               |
| PO2.  | <b>Problem Analysis:</b> Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)                                                                                                      |
| PO3.  | <b>Design/Development of Solutions:</b> Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5) |
| PO4.  | <b>Conduct Investigations of Complex Problems:</b> Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).                                                            |
| PO5.  | <b>Engineering Tool Usage:</b> Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)                                                                     |
| PO6.  | <b>The Engineer and The World:</b> Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).                                       |
| PO7.  | <b>Ethics:</b> Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)                                                                                                                                                       |
| PO8.  | <b>Individual and Collaborative Team work:</b> Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.                                                                                                                                                                   |
| PO9.  | <b>Communication:</b> Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences                      |
| PO10. | <b>Project Management and Finance:</b> Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.                                          |
| PO11. | <b>Life-Long Learning:</b> Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)                                                  |



### **Program Specific Outcomes (PSOs)**

|              |                                                                                                                                                                                                                           |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PSO1.</b> | <b>Problem-Solving Skills:</b> The ability to apply standard practices and strategies in software project development using open-ended programming environments to deliver a quality product for the benefit of students. |
| <b>PSO2.</b> | Design, Implement, Test and Evaluate a computer system, component or algorithm to meet desired needs and to solve a computational problem.                                                                                |



**STANLEY COLLEGE OF ENGINEERING & TECHNOLOGY FOR WOMEN**

(Autonomous)

Abids, Hyderabad - 500 001, Telangana

# *Department of Computer Science and Engineering*

## **Objectives and Outcomes of the Case Study**

### **Course Objectives:**

1. To get familiar with fundamental concepts of database managements and with database designing.
2. To master hands on SQL and PL/SQL concepts.
3. To impart knowledge in indexing, hashing, transaction management, concurrency control techniques and recovery techniques.

### **Course Outcomes:**

After completing this course, the student will be able to:

1. Understand the role of database management system in an organization and learn the database concepts.
2. Construct database queries using relational algebra and SQL.
3. Design databases using data modeling and logical database design techniques.
4. Evaluating the indexing, hashing techniques and transaction management.
5. Understand the concept of a database transaction and related concurrent, recovery facilities.



**STANLEY COLLEGE OF ENGINEERING & TECHNOLOGY FOR WOMEN**

(Autonomous)

Abids, Hyderabad - 500 001, Telangana

# Department of Computer Science and Engineering

## Mapping of Course Outcomes with POs and PSOs

| Course Outcomes (COs)                                                                                  | Program Outcomes (POs) |     |     |     |     |     |     |     |     |      |      | Program Specific Outcomes (PSOs) |      |
|--------------------------------------------------------------------------------------------------------|------------------------|-----|-----|-----|-----|-----|-----|-----|-----|------|------|----------------------------------|------|
|                                                                                                        | PO1                    | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1                             | PSO2 |
| 1. Understand the role of database management system in an organization & learn the database concepts. | 3                      | 2   | 1   | 1   | 1   | 1   | 1   | 1   | 1   | 1    | 2    | 2                                | 1    |
| 2. Construct database queries using relational algebra and SQL.                                        | 3                      | 3   | 2   | 1   | 3   | 1   | 1   | 1   | 1   | 1    | 2    | 3                                | 3    |
| 3. Design databases using data modeling and logical database design techniques.                        | 3                      | 3   | 3   | 2   | 2   | 1   | 1   | 1   | 1   | 1    | 2    | 3                                | 3    |
| 4. Evaluating the indexing, hashing techniques and transaction management.                             | 3                      | 3   | 2   | 2   | 3   | 1   | 1   | 1   | 1   | 1    | 2    | 3                                | 3    |
| 5. Understand the concept of a database transaction and related concurrent, recovery facilities.       | 3                      | 3   | 2   | 2   | 3   | 1   | 1   | 1   | 1   | 1    | 2    | 2                                | 3    |



**STANLEY COLLEGE OF ENGINEERING & TECHNOLOGY FOR WOMEN**

(Autonomous)

Abids, Hyderabad - 500 001, Telangana

# *Department of Computer Science and Engineering*

## **Student Learning Outcomes**

- Skills acquired (technical, analytical, communication, teamwork, etc.)
- Tools / technologies learned
- Industry exposure and professional ethics



**STANLEY COLLEGE OF ENGINEERING & TECHNOLOGY FOR WOMEN**

(Autonomous)

Abids, Hyderabad - 500 001, Telangana

# Department of Computer Science and Engineering

## Mapping of Internship with SDGs

**Roll No:**

**Name:**

**Class, Semester & Section:**

**Academic Year:**

**Title of Case Study:**

**Domain:**

**Course Outcomes (COs) Mapping with SDGs:**

### Sample SDG Mapping with Justification

| Course Outcomes (COs)                                                                                             | Relevant SDGs                                          | Justification of SDGs                                                                                                                                             |
|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CO1:</b> Understand the role of database management system in an organization and learn the database concepts. | <b>SDG 4 - Quality Education</b>                       | Learning DBMS concepts enhances technical knowledge, problem-solving ability, and digital literacy required for modern education and computing fields.            |
| <b>CO2:</b> Construct database queries using relational algebra and SQL                                           | <b>SDG 9 - Industry, Innovation and Infrastructure</b> | SQL and query-building skills support the development of efficient digital systems and modern technological infrastructure used in industries.                    |
| <b>CO3:</b> Design database queries using data modeling and logical database design techniques.                   | <b>SDG 9 - Industry, Innovation and Infrastructure</b> | Database design contributes to innovative, scalable, and reliable software solutions that strengthen information infrastructure.                                  |
| <b>CO4:</b> Evaluating the indexing, hashing techniques and transaction management.                               | <b>SDG 8 - Decent Work and Economic Growth</b>         | Efficient database management techniques improve system performance and productivity, supporting organizational growth and technological efficiency.              |
| <b>CO5:</b> Understand the concept of a database transaction and related concurrent, recovery facilities.         | <b>SDG 16 - Peace, Justice and Strong Institutions</b> | Transaction management and recovery mechanisms ensure secure, reliable, and consistent data systems that support trustworthy digital operations and institutions. |

### **Sample Mapping**

| <b>Internship Domain</b>               | <b>Relevant SDGs</b>                                                   | <b>Justification</b>                                                    |
|----------------------------------------|------------------------------------------------------------------------|-------------------------------------------------------------------------|
| Web / App Development                  | SDG 9 - Industry, Innovation and Infrastructure                        | Developed scalable web solutions improving industrial workflows.        |
| Artificial Intelligence / Data Science | SDG 4 - Quality Education, SDG 11 - Sustainable Cities and Communities | Applied machine learning for predictive analytics aiding smart systems. |
| Cybersecurity                          | SDG 16 - Peace, Justice and Strong Institutions                        | Worked on security compliance and ethical data protection practices.    |
| Cloud Computing / DevOps               | SDG 8 - Decent Work and Economic Growth                                | Implemented efficient, automated cloud-based deployment systems.        |

## ACKNOWLEDGEMENT

The satisfaction that accompanies the successful completion of the task would be put incomplete without the mention of the people who made it possible, whose constant guidance and encouragement crown all the efforts with success.

The college and our mentor Dr. P. R. Anisha are very helpful and continuously encourages us to complete the task assigned by the department. College support is also very good. They provided all the basic requirements required to complete the task.

We express our heartfelt thanks to **Dr. P. R. Anisha**, Associate Professor CSE, for her suggestions in the selection and carrying out an in-depth study of the topic. Her valuable guidance and encouragement really helped us to shape this report to perfection.

We wish to express our deep sense of gratitude to **Dr. R. Madana Mohana**, Professor & Head, for his intense support and encouragement, which helped us to mold our internship into a successful one.

We also owe our special thanks to our honorable Principal **Dr. B. L. Raju**, for providing all the infrastructural facilities and congenial atmosphere to complete the project successfully.

We also thank all the staff members of the Computer Science and E department for their valuable support and generous advice.

Finally, thanks to all my friends and family members for their continuous support and enthusiastic help.

1. **Kausar Jehan**  
**(160624733156)**
2. **Umama Nooreen**  
**(160624733188)**

## ABSTRACT

(Brief summary about 150-200 words describing the case study work, objectives, and outcomes.)

The Traffic Violation Management System is a database-driven application developed using MySQL to efficiently manage traffic violation records and vehicle-related information. The primary objective of this case study is to apply Database Management System concepts to solve real-world problems associated with manual traffic record maintenance.

The system stores details of vehicle owners, registered vehicles, traffic violation rules, and records of violations. It uses relational database concepts such as primary keys, foreign keys, constraints, and views to ensure data integrity and efficient data retrieval. SQL operations including Data Definition Language (DDL), Data Manipulation Language (DML), and queries were implemented to create and manage the database.

A view named *ViolationHistory* was created to display comprehensive violation information, while SQL queries were used to identify repeat offenders and analyze traffic records. The project demonstrates the practical application of database design, normalization, and query processing techniques.

This case study enhances understanding of DBMS concepts and showcases how database technologies can contribute to safer roads, efficient law enforcement, and smart city initiatives aligned with sustainable development goals.



## Table of Contents

|                                                                    | Page No's    |
|--------------------------------------------------------------------|--------------|
| <b>Title Page</b>                                                  | <b>I</b>     |
| <b>College Certificate</b>                                         | <b>II</b>    |
| <b>Declaration</b>                                                 | <b>III</b>   |
| <b>Case Study Details</b>                                          | <b>IV</b>    |
| <b>Resume of Team Members</b>                                      | <b>V</b>     |
| <b>Vision, Mission and Program Educational Objectives (PEOs)</b>   | <b>VI</b>    |
| <b>Knowledge and Attitude Profile (WK)</b>                         | <b>VII</b>   |
| <b>Program Outcomes (POs) and Program Specific Outcomes (PSOs)</b> | <b>VIII</b>  |
| <b>Objectives and Outcomes of the Case Study</b>                   | <b>X</b>     |
| <b>Course Outcomes mapping with CO-PO/PSO</b>                      | <b>XII</b>   |
| <b>SDGs mapping with Justification</b>                             | <b>XIV</b>   |
| <b>Acknowledgement</b>                                             | <b>XV</b>    |
| <b>Abstract</b>                                                    | <b>XVI</b>   |
| <b>Table of Contents</b>                                           | <b>XVII</b>  |
| <b>List of Figures</b>                                             | <b>XVIII</b> |
| <b>List of Tables</b>                                              | <b>XIX</b>   |
| <b>Symbols &amp; Abbreviations</b>                                 | <b>XX</b>    |
| <b>WORK / PROJECT DESCRIPTION</b>                                  |              |
| <b>1. Problem statement</b>                                        | <b>1</b>     |
| <b>2. SDG mapping</b>                                              | <b>2</b>     |
| <b>3. GitHub Repository</b>                                        | <b>3</b>     |
| <b>4. Solution for the problem</b>                                 | <b>4</b>     |
| <b>5. Collaboration by the team</b>                                | <b>5</b>     |
| <b>6. Software development cycle</b>                               | <b>6</b>     |
| <b>7. Conclusion and Future Scope</b>                              | <b>7</b>     |
| <b>8. References</b>                                               | <b>8</b>     |

## List of Figures

| Sample Data |                  |          |
|-------------|------------------|----------|
| FIGURE NO.  | TITLE OF FIGURES | PAGE NO. |
| 1           | ER Diagram       | 30       |

## List of Tables

Sample Data

| <b>Table no</b> | <b>TITLE OF Table</b> | <b>PAGE NO.</b> |
|-----------------|-----------------------|-----------------|
| <b>1</b>        | Vehicle owners        | <b>30</b>       |
| <b>2</b>        | vehicles              | <b>30</b>       |
| <b>3</b>        | Traffic violations    | <b>30</b>       |
| <b>4</b>        | Violation entry       | <b>31</b>       |
| <b>5</b>        | Violation history     | <b>31</b>       |

## Symbols & Abbreviations

Sample Data

**Abbreviation**

**Meaning**

**PAGE NO.**

## Project Description/SDG Mapping/Github

(Explain about the case study, the sdg mapping and github link)

### Project Description

The Traffic Violation Management System is a database application developed using MySQL to efficiently manage traffic violation records and related information. Traditional traffic management systems often rely on manual record keeping, which can lead to errors, data redundancy, and difficulty in tracking offenders. This project aims to provide a structured and automated approach to managing traffic-related data.

The system consists of four main entities: VehicleOwner, Vehicle, TrafficViolation, and ViolationEntry. The VehicleOwner table stores information about vehicle owners, while the Vehicle table maintains details of registered vehicles. The TrafficViolation table contains information regarding traffic rules and corresponding fines, and the ViolationEntry table records each violation committed by vehicles.

The database is implemented using SQL with various constraints such as PRIMARY KEY, FOREIGN KEY, UNIQUE, NOT NULL, CHECK, and AUTO\_INCREMENT to ensure data integrity and consistency. A database view named *ViolationHistory* has been created to simplify retrieval of violation details. The project also includes SQL queries to display records and identify repeat offenders.

---

The project aligns with the following United Nations Sustainable Development Goals (SDGs):

### SDG 3: Good Health and Well-Being

The system promotes road safety by maintaining accurate traffic violation records and encouraging adherence to traffic regulations, thereby helping reduce road accidents.

### SDG 9: Industry, Innovation and Infrastructure

The project utilizes database technologies to create efficient digital infrastructure and demonstrates innovative use of information systems.

### SDG 11: Sustainable Cities and Communities

Efficient traffic management contributes to safer and smarter cities through improved transportation governance.

### SDG 16: Peace, Justice and Strong Institutions

The system ensures transparent and reliable record management, supporting fair enforcement of traffic laws.

---

### GitHub Repository

The source code, SQL scripts, documentation, screenshots, and project files are maintained using Git and GitHub for version control and collaboration.

### GitHub Repository Link:

<https://github.com/kausar-296/traffic-violation-management-system.git>

The repository contains:

- Database schema files
- SQL scripts
- Sample data
- Query implementations
- Screenshots of execution
- Project documentation
- README file

Git and GitHub were used to track changes, maintain project history, and enable collaboration among team members.

## Solving the complex engineering problem

(explain about the complex engineering problem and your solution (in words) in the document and in the github link, the codes/deployment link/working solution of the solution)

Traffic management in modern cities is a complex engineering problem due to the large volume of vehicles, increasing traffic violations, and the need for efficient law enforcement. Manual maintenance of traffic records often results in data redundancy, inconsistency, difficulty in tracking offenders, and inefficient retrieval of information. Identifying repeat offenders and maintaining accurate records become increasingly challenging as the amount of data grows.

To address this problem, the Traffic Violation Management System was developed using MySQL. The proposed solution employs a relational database model to efficiently organize and manage traffic-related information. Four interrelated tables—VehicleOwner, Vehicle, TrafficViolation, and ViolationEntry—were designed using normalization principles to reduce redundancy and maintain data integrity.

The system uses various database constraints such as PRIMARY KEY, FOREIGN KEY, UNIQUE, NOT NULL, CHECK, and AUTO\_INCREMENT to ensure consistency and reliability of data. SQL queries were implemented to insert, retrieve, and analyze information efficiently. A database view named *ViolationHistory* was created to provide a consolidated view of vehicle number plates, violation rules, and timestamps of violations.

The system also supports identification of repeat offenders through aggregate SQL queries, thereby assisting authorities in monitoring habitual violators. The use of relational database concepts enables efficient data storage, retrieval, and management, making the system scalable and reliable for real-world applications.

## GitHub Repository

<https://github.com/kausar-296/traffic-violation-management-system.git>

The repository contains:

- SQL source code
- Database schema
- Sample data
- Query implementations
- Screenshots of execution
- Documentation and README file

## Working Solution

The developed system successfully performs the following operations:

1. Creation of database and tables.
2. Insertion of vehicle owner, vehicle, and violation data.

3. Enforcement of integrity constraints.
4. Retrieval of records using SQL queries.
5. Display of violation history through database views.
6. Identification of repeat offenders.
7. Maintenance of structured and normalized data.

The solution demonstrates the practical application of Database Management System concepts in solving real-world traffic management challenges and contributes towards safer and smarter transportation systems.



## Elaborate about the Collaboration

(Which members did what part)

The Traffic Violation Management System was developed collaboratively by the team members, with responsibilities divided according to individual strengths and interests. Both members actively participated in the planning, design, implementation, testing, and documentation phases of the project.

Kausar Jehan:

Responsibilities:

- Identified the problem statement and project objectives.
- Designed the database schema and entity relationships.
- Created SQL scripts for database and table creation.
- Implemented constraints such as PRIMARY KEY, FOREIGN KEY, UNIQUE, NOT NULL, CHECK, and AUTO\_INCREMENT.
- Created SQL queries and database views.
- Set up Git and GitHub repository for version control.
- Managed commits, updates, and project organization.

Umama Nooreen:

Responsibilities:

- Prepared sample datasets for testing.
- Verified query outputs and database functionality.
- Assisted in testing and debugging SQL statements.
- Prepared screenshots of execution and outputs.
- Created the ER Diagram and supporting documentation.
- Contributed to report writing and formatting.

Both team members jointly:

- Discussed project requirements and scope.
- Designed the Traffic Violation Management System.
- Reviewed database normalization and relationships.
- Tested SQL queries and verified results.
- Maintained project files using GitHub.
- Prepared the final report and presentation.

The collaborative approach improved teamwork, communication skills, and project management abilities while ensuring successful completion of the case study.

## Software Development Cycle

The development of the Traffic Violation Management System followed the Software Development Life Cycle (SDLC) to ensure systematic planning, implementation, and testing of the project.

### 1. Requirement Analysis

The first phase involved identifying the problems associated with manual traffic record management. The requirements of the system were analyzed, including storage of vehicle details, owner information, traffic violations, and violation records.

### 2. System Design

Based on the requirements, the database schema was designed. Four entities—VehicleOwner, Vehicle, TrafficViolation, and ViolationEntry—were identified. Relationships between entities were established using primary keys and foreign keys. An Entity Relationship (ER) diagram was prepared to represent the database structure.

### 3. Implementation

The database was implemented using MySQL. SQL scripts were written to create the database and tables. Constraints such as PRIMARY KEY, FOREIGN KEY, UNIQUE, NOT NULL, CHECK, and AUTO\_INCREMENT were applied to maintain data integrity. Sample records were inserted into the tables using SQL INSERT statements.

### 4. Testing

The developed system was tested by executing SQL queries to verify data insertion, retrieval, and integrity constraints. Views and queries were tested to ensure accurate outputs. Sample data was used to validate system functionality and identify repeat offenders.

### 5. Deployment and Version Control

The project files, SQL scripts, screenshots, and documentation were maintained using Git and GitHub. Version control was used to track changes, manage collaboration among team members, and maintain project history.

GitHub Repository:

<https://github.com/kausar-296/traffic-violation-management-system.git>

## 6. Maintenance and Future Enhancements

The system can be further enhanced by integrating features such as online fine payment, SMS and email notifications, police dashboards, mobile applications, and automatic violation detection using CCTV cameras.

Thus, by following the Software Development Life Cycle, the project was developed in a structured and efficient manner while ensuring reliability, maintainability, and scalability.

## Conclusion and Future Scope

(Summarize the overall outcome of your internship)

The Traffic Violation Management System was successfully designed and implemented using MySQL as a relational database management system. The project effectively demonstrates the practical application of Database Management System concepts such as database design, normalization, SQL queries, constraints, and views in solving a real-world problem.

The developed system efficiently stores and manages information related to vehicle owners, vehicles, traffic violations, and violation records. By using PRIMARY KEY, FOREIGN KEY, UNIQUE, NOT NULL, CHECK, and AUTO\_INCREMENT constraints, data integrity and consistency were maintained throughout the database. The creation of the *ViolationHistory* view enabled efficient retrieval of violation information, while SQL queries facilitated analysis of traffic records and identification of repeat offenders.

The project enhanced our understanding of database modeling, SQL implementation, version control using Git and GitHub, and collaborative software development. It also improved our problem-solving, analytical, teamwork, and documentation skills.

## Future Scope

The Traffic Violation Management System can be further enhanced with additional features and technologies to make it suitable for real-world deployment. Some possible future enhancements include:

- Integration of an online fine payment system.
- Development of a graphical user interface (GUI) or web application.
- SMS and email notifications for traffic violations.
- Police administration dashboard for monitoring records.
- Mobile application support for users and authorities.
- Automatic violation detection using CCTV cameras and Artificial Intelligence.
- Real-time analytics and reporting dashboards.
- Generation of e-challans and digital receipts.

Thus, the project provides a strong foundation for developing intelligent and efficient traffic management systems that contribute towards safer roads and smart cities.

## References

(List any references, tools, or materials used.)

### SAMPLE FORMAT

[1]GeeksforGeeks SQL Tutorial. Available at: <https://www.geeksforgeeks.org/sql-tutorial/>

[2]Visual Studio Code Documentation. Microsoft Corporation. Available at: <https://code.visualstudio.com/docs>

[3]Elmasri, R., & Navathe, S. B. *Fundamentals of Database Systems* (7th Edition). Pearson Education.

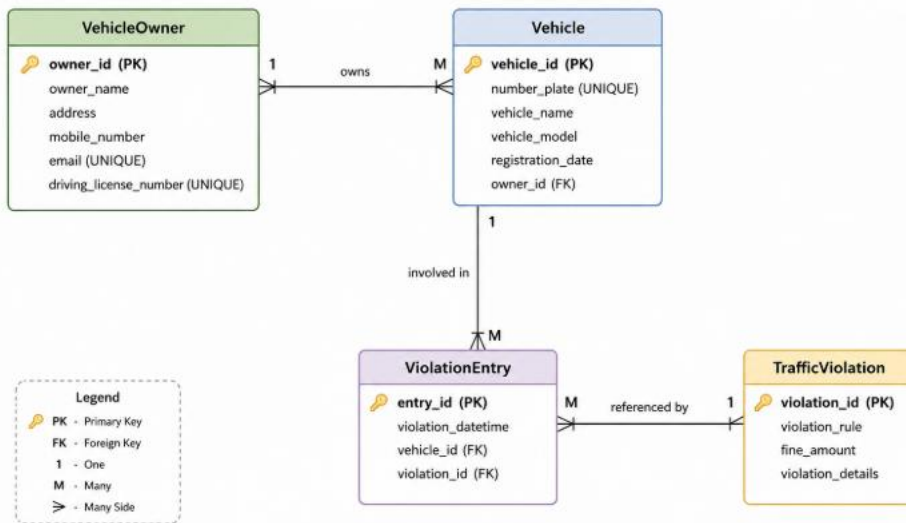
### Tools and Technologies Used

- MySQL 8.0
- MySQL Command Line Client
- Visual Studio Code
- Git
- GitHub
- Microsoft Word

### Materials Used

- DBMS course materials and lecture notes
- SQL syntax reference guides
- MySQL documentation and tutorials
- Git and GitHub documentation

## ER Diagram - Traffic Violation Management System



```
mysql> SELECT * FROM VehicleOwner;
```

| owner_id | owner_name     | address   | mobile_number | email             | driving_license_number |
|----------|----------------|-----------|---------------|-------------------|------------------------|
| 1        | Rahul Sharma   | Hyderabad | 9876543210    | rahul@gmail.com   | DL001                  |
| 2        | Saifullah Khan | Delhi     | 9876543211    | saif@gmail.com    | DL002                  |
| 3        | John Anthony   | Mumbai    | 9878375210    | john02@gmail.com  | DL003                  |
| 4        | Sabrina        | Hyderabad | 9876593850    | rina934@gmail.com | DL004                  |

```
4 rows in set (0.04 sec)
```

```
mysql> SELECT * FROM Vehicle;
```

| vehicle_id | number_plate | vehicle_name | vehicle_model | registration_date | owner_id |
|------------|--------------|--------------|---------------|-------------------|----------|
| 1          | TS09AB1234   | Aktiva       | 2022          | 2022-06-15        | 1        |
| 2          | DL01XY5678   | Swift        | 2021          | 2021-08-20        | 2        |
| 3          | MH12CD4321   | Pulsar       | 2023          | 2023-02-10        | 3        |
| 4          | TS08XY9876   | Creta        | 2024          | 2024-01-15        | 4        |

```
4 rows in set (0.04 sec)
```

```
mysql> SELECT * FROM TrafficViolation;
```

| violation_id | violation_rule | fine_amount | violation_details        |
|--------------|----------------|-------------|--------------------------|
| 1            | No Helmet      | 500         | Rider not wearing helmet |
| 2            | Signal Jump    | 1000        | Crossed red signal       |
| 3            | Over Speeding  | 1500        | Exceeded speed limit     |

```
3 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM ViolationEntry;
```

| entry_id | violation_datetime  | vehicle_id | violation_id |
|----------|---------------------|------------|--------------|
| 1        | 2026-06-05 10:30:00 | 1          | 1            |
| 2        | 2026-06-05 14:45:00 | 2          | 2            |
| 3        | 2026-06-06 09:10:00 | 2          | 3            |
| 4        | 2026-06-06 16:30:00 | 2          | 1            |

```
4 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM ViolationHistory;
```

| number_plate | violation_rule | violation_datetime  |
|--------------|----------------|---------------------|
| TS09AB1234   | No Helmet      | 2026-06-05 10:30:00 |
| DL01XY5678   | No Helmet      | 2026-06-06 16:30:00 |
| DL01XY5678   | Signal Jump    | 2026-06-05 14:45:00 |
| DL01XY5678   | Over Speeding  | 2026-06-06 09:10:00 |

```
4 rows in set (0.04 sec)
```